VARIANCE-WEIGHTED RESIDUAL SUBSPACES FOR MISSING ION-RESOLVED READOUTS

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Abstract. Ion-resolved experiments produce short, heteroscedastic sequences for which a missing detector interval can remove a substantial fraction of one spatial profile. A residual-subspace reconstruction is introduced for this setting. Complete development profiles are centered separately, their leading right singular vectors define a spatial residual basis, and an incomplete profile is reconstructed by variance-weighted least squares in the span of that basis and the constant vector. A deterministic theorem separates unmodeled residual on missing coordinates from noise amplified through the sampled basis. Exact recovery follows when the profile lies in the learned span and the sampled basis has full column rank. Recovery costs $O(n_o r^2 + r^3 + nr)$ operations and stores nr basis scalars. Evaluation uses 561 authenticated site-resolved averages from a 51-ion experiment, each derived from 1000 shots. Seven early times select $r = 4$ by leave-one-time-out reconstruction; times 8–10 ms remain untouched for confirmation. Across periodic one-in-five losses and five contiguous ten-ion outages, confirmation RMSE is 0.06270, compared with 0.08435 for the strongest simple comparator. This is a 44.7 percent reduction in pooled squared error. The method wins 25 of 30 mask–time cases; all five losses and the complete rank sensitivity are reported. The result supports a compact readout-recovery primitive, not universal dominance or a microscopic dynamical model.

1. Ion-resolved recovery problem. Consider a measured profile $x \in \mathbb{R}^n$ with observed coordinate set Ω and missing set Γ. Coordinate restrictions are denoted by P_Ω and P_Γ. The data in this study are staggered-spin averages from a 51-ion Néel-state experiment [1, 2]. The archive contains eleven times and 51 sites, with a reported standard error for every average. Missing-coordinate recovery differs from forecasting: measurements at the retained sites of the current time are available, while a periodic detector loss or contiguous outage hides the remaining sites.

Spatial interpolation assumes local smoothness. That assumption is poorly matched to the alternating and boundary-dependent residuals in the measured profiles. A constant fill discards persistent spatial structure. The method below retains a constant component for the instantaneous global level and learns only the spatial residual directions from earlier complete profiles. This separation prevents the dominant temporal oscillation from consuming the limited rank.

2. Variance-weighted residual subspace. Let $X \in \mathbb{R}^{T\times n}$ contain complete development profiles. Define

$$R = X - \operatorname{diag}(X\mathbf{1}/n)\mathbf{1}^T, \qquad R = Q\Sigma V^T, \tag{2.1}$$

and let $U = V_{:,1:r}$. The reconstruction design is $A = [\mathbf{1}\ U] \in \mathbb{R}^{n\times(r+1)}$. If s_i is the reported standard error, the observed-coordinate weight is $W = \operatorname{diag}(1/\max\{s_i, 10^{-2}\})_{i\in\Omega}$. The coefficient and reconstructed profile are

$$\widehat{z} = \arg\min_{z\in\mathbb{R}^{r+1}} \|W(P_\Omega A z - P_\Omega x)\|_2^2 + \lambda \|z_{2:r+1}\|_2^2, \tag{2.2}$$

$$\widehat{x} = A\widehat{z}. \tag{2.3}$$

The registered grid contains $r = 1, \dots, 7$ and $\lambda = 0$; the latter is fixed because every development sampled design is well conditioned at the candidate ranks. Ties are resolved toward smaller rank.

Theorem 1 (Missing-coordinate error)**.** *Assume $\lambda = 0$ and $M = WP_\Omega A$ has full column rank. For every $x = Az_\star + e$,*

$$\|P_\Gamma(\widehat{x} - x)\|_2 \le \|P_\Gamma e\|_2 + \frac{\|P_\Gamma A\|_2}{\sigma_{\min}(M)} \|WP_\Omega e\|_2 . \tag{2.4}$$

In particular, $e = 0$ implies $\widehat{x} = x$.

Proof. Substitution of $P_\Omega x = P_\Omega A z_\star + P_\Omega e$ into the full-rank weighted least-squares solution gives

$$\widehat{z} = M^\dagger W P_\Omega x = z_\star + M^\dagger W P_\Omega e. \tag{2.5}$$

Consequently,

$$P_\Gamma(\widehat{x} - x) = P_\Gamma A M^\dagger W P_\Omega e - P_\Gamma e. \tag{2.6}$$

The triangle inequality, submultiplicativity, and $\left\|M^\dagger\right\|_2 = 1/\sigma_{\min}(M)$ prove Eq. (2.4). When $e = 0$, Eq. (2.5) gives $\widehat{z} = z_\star$ and hence exact recovery. □

The two terms in Eq. (2.4) expose distinct limitations. The first is residual structure absent from the learned span on the missing sites. The second is observed residual amplified by poor sampling of the basis. Thus a larger rank can reduce approximation error while decreasing $\sigma_{\min}(M)$, an effect measured directly in the rank experiment.

Theorem 2 (Arithmetic and storage)**.** *Given U, one reconstruction with $n_o = |\Omega|$ and rank r uses $O(n_o r^2 + r^3 + nr)$ arithmetic and $O(nr + r^2)$ auxiliary storage. The stored model contains exactly nr basis scalars in addition to singular values.*

Proof. Forming the weighted normal matrix and right-hand side costs $O(n_o r^2)$. Factoring the $(r+1)$-dimensional normal matrix costs $O(r^3)$, and applying A costs $O(nr)$. The basis and normal matrix occupy nr and $O(r^2)$ scalars, respectively. No $n \times n$ matrix is formed. □

3. Registered experimental design. The source archive and license are recorded in Table 1. Times 1–7 ms form development cases. Each development time is omitted in turn from the basis construction, preventing its hidden entries from influencing rank selection. After selecting the rank, times 0–7 ms form the fixed basis for confirmation at 8–10 ms. The two mask families contain five offsets each: periodic removal of one site in five and contiguous removal of ten sites. The resulting confirmation set contains 30 mask–time cases.

Comparators use only the retained coordinates of the current profile. They are the observed-site mean, piecewise-linear spatial interpolation, and nearest-site fill. RMSE is computed only on hidden coordinates. Pooled RMSE is the square root of mean squared error over the registered cases; paired wins are counted at the mask–time level.

4. Results. Figure 1 shows the complete rank path. The minimum of the development curve occurs at $r = 4$. Confirmation also remains below the strongest simple comparator throughout the registered rank range, while its minimum at rank two is not used for model selection.

Table 2 gives the registered aggregate. The improvement is not uniform. Figure 2(a) displays every confirmation case and marks each loss, preventing the pooled statistic from hiding five adverse cases. The contiguous 21–30 outage at 10 ms in Fig. 2(b) illustrates the mechanism: retained sites determine the instantaneous offset and four development residual directions interpolate a nonlocal spatial pattern.

Figure 1(b) connects the measured cost to Theorem 2. Storage grows exactly as $51r$ scalars. The small timing scale reflects this data set; the operation count, rather than a single-device measurement, determines extrapolation to larger arrays.

5. Scope. Residual-subspace recovery exploits repeated spatial structure but does not identify a Hamiltonian, Lindbladian, or measurement-noise mechanism. The experiment contains one 51-ion sequence and three confirmation times. Additional devices, calibrations, and independently acquired sequences are required before attributing broad physical generality. Within the fixed protocol, the main

Table 1: Authenticated data and fixed evaluation protocol.

Quantity	Value
Source	Zenodo 10.5281/zenodo.17750419
License	CC BY 4.0
Ion-resolved averages	561
Shots per time	1000
Development times	1–7 ms
Confirmation times	8–10 ms
Candidate ranks	1–7
Selected rank	4
Confirmation cases	30

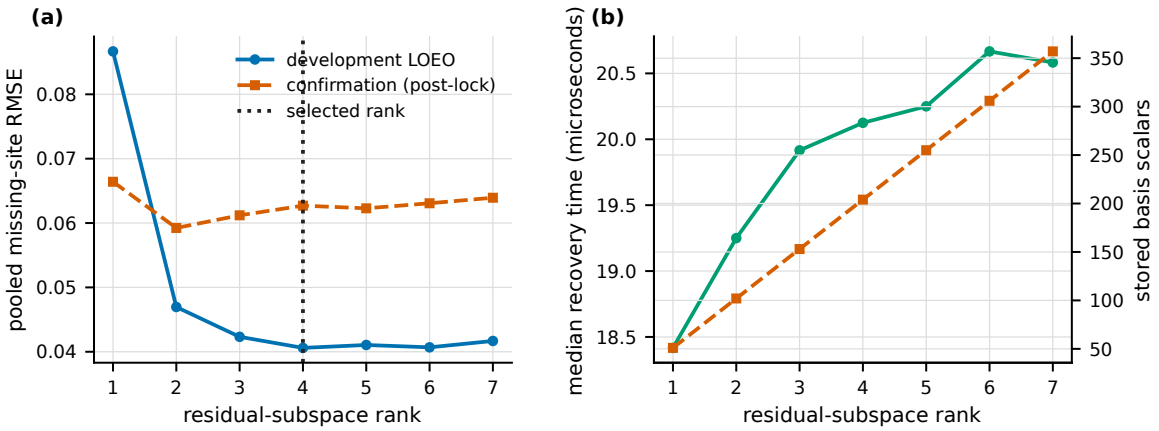


Fig. 1: Statistical and computational rank paths. (a) Early-time leave-one-out selection and post-lock confirmation sensitivity. Rank four minimizes the development curve; the confirmation curve is not a selection criterion. (b) Measured recovery time and exact basis storage. Timings are medians of 101 runs on the stated local CPU environment.

Table 2: Pooled confirmation error on 30 mask–time cases. The squared-error reduction is measured against the observed-site mean, the strongest simple comparator.

Method	Missing-site RMSE
Residual subspace, $r = 4$	0.06270
Observed-site mean	0.08435
Linear interpolation	0.12071
Nearest-site fill	0.12424
Squared-error reduction	44.7%
Paired wins vs. mean	25 of 30

empirical conclusion is narrower: early-time residual directions improve pooled missing-site recovery over three simple spatial fills, with five explicitly reported confirmation losses.

The implementation stores no square spatial operator. Every result table and plot is regenerated from the attributed CSV records, the registered masks, and a deterministic rank-selection rule. The theorem supplies the associated conditioning diagnosis: missing residual energy and sampled-basis conditioning, rather than visual smoothness, govern reconstruction accuracy.

References.

[1] Manoj Kumar Joshi, Florian Kranzl, Johannes Franke, Christian F. Roos, and Rainer Blatt. Bounded-error quantum simulation via hamiltonian and lindbladian learning, 2025.

[2] Tristan Kraft, Manoj Kumar Joshi, William Lam, Tobias Olsacher, Florian Kranzl, Johannes Franke, Lata Kh Joshi, Rainer Blatt, Augusto Smerzi, Daniel Stilck França, Benoît Vermersch, Barbara Kraus, Christian F. Roos, and Peter Zoller. Bounded-error quantum simulation via hamiltonian and lindbladian learning, 2025.

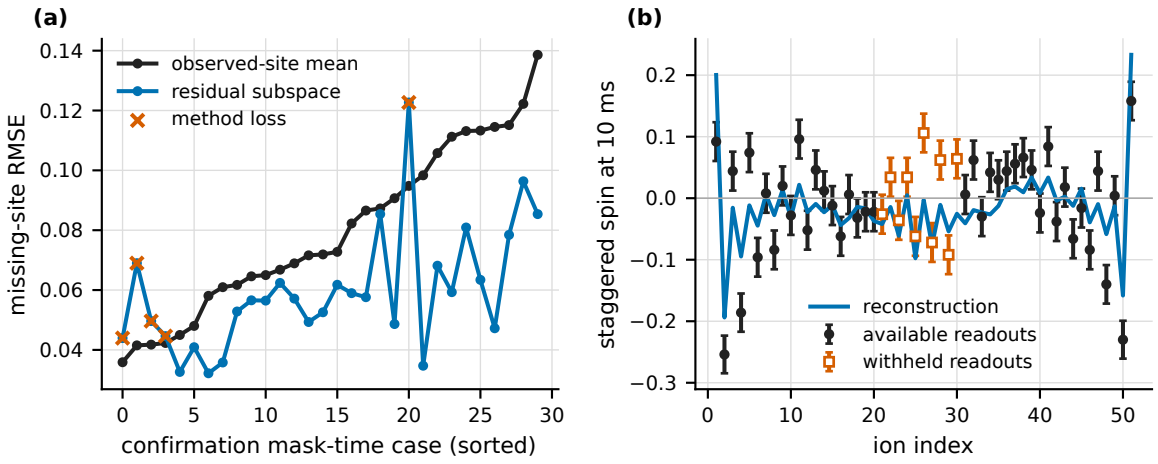


Fig. 2: Confirmation evidence. (a) All 30 cases, ordered by observed-mean error; crosses identify the five method losses. (b) A registered ten-site contiguous outage at 10 ms. Filled circles are available measurements, open squares are withheld measurements, and the line is the reconstruction. Error bars are reported standard errors.